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Project:Part 2

A.

Outlet:

This relation stores information about each rental outlet and the primary key is outlet_number

Vehicle:

The Vehicle relation stores all details of vehicles available for hire and the primary key here is registration_number. It also has Outlet_number as foreign key that shows which outlet the vehicle belongs to.

Client:

Client relation stores the customer details and the primary key is client_number.

Staff:

Staff stores information about employees working at different outlets and the primary key is staff_number which also includes outlet_number as foreign key to show which outlet the staff member is assigned to.

Hire Agreement:

This represents each rental transaction and the primary key here is hire_number which also includes the client_number as foreign key referencing the client and registration_number as foreign key for vehicles.

Relational Schema:

Outlet(Outlet_number, Address, Phone_number, Fax_number)

Vehicle(Registration_number, Make, Model, Engine_size, Capacity, Current_mileage, Daily_hire_rate, Outlet_number)

Client(Client_number, First_name, Last_name, Address, Phone_number, DOB, Driver_license_number)

Staff(Staff_number, First_name, Last_name, Home_address, Home_phone_number, DOB, Sex, Date_joined, Job_title, Salary, Outlet_number)

Hire_Agreement(Hire_number, Date_started, Date_wish_end, Mileage_before, Mileage_after, Client_number, Registration_number)

B.

Outlet(Outlet_number, Address, Phone_number, Fax_number)

1 NF: All attributes hold to a single value. No repeating groups exist.

2 NF: The primary key here is Outlet_Number and it is a single attribute which makes partial dependency impossible

3 NF: Address, Phone Number and Fax Number are all directly describe the outlet and which says none depend on each other.

Vehicle(Registration_number, Make, Model, Engine_size, Capacity, Current_mileage, Daily_hire_rate, Outlet_number)

1 NF: All attributes are atomic and no repeating groups exist.

2 NF: The primary key here is Registration_Number and again it's a single attribute, so partial dependencies are impossible.

3 NF: There is a possibility that the Model could determine Make but we assume here each of them are recorded per individual not derived from each other. All attributes describe directly to the vehicle which means no transitive dependencies exist.

Assumption: Make and Model are treated as independent attributes, not a separate lookup table.

Client(Client_number, First_name, Last_name, Address, Phone_number, DOB, Driver_license_number)

1 NF: All attributes are atomic with no repeating groups.

2 NF: The primary key is the client number and again it's a single attribute which makes partial dependencies impossible.

3 NF: All attributes describe the client. No transitive dependencies exist.

Staff(Staff_number, First_name, Last_name, Home_address, Home_phone_number, DOB, Sex, Date_joined, Job_title, Salary, Outlet_number)

1 NF: All attributes are atomic with no repeating groups.

2 NF: The primary key is Staff_number, a single attribute, so partial dependencies are impossible.

3 NF: A potential transitive dependency exists between Job_title and Salary, salary is determined by job title which could suggest a separate Job table. But, we assume salary is assigned individually per staff member and may vary person to person regardless of title. No transitive dependencies exist under this assumption.

Assumption: Salary is an individual attribute per staff member, not determined solely by job title.

Hire_Agreement(Hire_number, Date_started, Date_wish_end, Mileage_before, Mileage_after, Client_number, Registration_number)

1 NF: All attributes are atomic with no repeating groups.

2 NF: The primary key is Hire_number and a single attribute which makes partial dependency impossible.

3 NF: This table is really important to validate. Storing client name, address, phone number, model, make directly on the hire agreement. These would create transitive dependency on the client_number and registration_number, not on the hire_number. Our schema here eliminates this redundancy by storing only the foreign keys, and retrieving client and vehicle information from their respective tables when needed which means no transitive dependencies exist.

All the five relations: Outlet, Vehicle, Client, Staff, and Hire_Agreement satisfy 1NF, 2NF, and 3NF.

C.

1. List all of the vehicles owned by a single client.
 - a. Use client_number to identify the single client being referenced.
 - b. Locate all Hire_agreement (Hire_number = PK) specific to the identified client.
 - c. Within each Hire_agreement/hire_number to identify all vehicles owned by the single client.
2. List all of the vehicles above 30,000 miles that were hired by clients during 1/1/25-12/31/25.
 - a. Identify vehicles whose current_mileage was above 30,000
 - b. Identify what vehicles have been hired using Registration_number between Vehicles and Hire_Agreement.
 - c. Filter and select record where date_started falls between 1/1/25-12/31/25.
 - d. Confirm that each of the selected vehicles was hired by clients.
 - e. After filtering, find the sum of the relative vehicles through registration_numbers
3. List the total amount of Hondas amongst all outlets.
 - a. Identify the total number of vehicles with the Make attribute of "Honda".
 - b. Identify what outlets host each "Honda"
 - c. Identify Hondas across all outlets, not a specific one.
 - d. Run count of all Hondas within all outlets.
4. List the vehicle Make that cost the most to hire (daily_hire_rate).
 - a. Identify the daily_hire_rate for all vehicles.
 - b. Filter to identify which vehicles have the highest daily_hire_rate
 - c. Confirm that the vehicles exist in an outlet.
 - d. With the filtered vehicles we find the Make.
 - e. Retrieve the final output with the vehicle makes that have the highest daily_hire_rate.
5. Show the 10 outlets have the most staff.
 - a. Each staff member is linked to an outlet so we connect each staff to the outlet they are employed.
 - b. Organize staff to their associated outlet.

- c. Run a count of the amount of staff at each outlet.
- d. Rearrange the outlets based on the amount of staff they have, highest to lowest.
- e. Filter out the highest 10 outlets with the most staff.

D. Integrity Constraints

D.1 Primary Key Constraints

The following primary keys uniquely identify each tuple in their respective relations:

Outlet

Primary Key: Outlet_number

Vehicle

Primary Key: Registration_number

Client

Primary Key: Client_number

Staff

Primary Key: Staff_number

Hire_Agreement

Primary Key: Hire_number

D.2 Referential Integrity / Foreign Key Constraints

The following foreign keys are required to represent the relationships between entities in the logical model:

Vehicle

Outlet_number is a foreign key referencing Outlet(Outlet_number)

Staff

Outlet_number is a foreign key referencing Outlet(Outlet_number)

Hire_Agreement

Client_number is a foreign key referencing Client(Client_number)

Registration_number is a foreign key referencing Vehicle(Registration_number)

These constraints ensure that:

A vehicle must belong to an existing outlet

A staff member must be assigned to an existing outlet

A hire agreement must refer to an existing client

A hire agreement must refer to an existing vehicle

D.3 Alternate Key Constraints

An alternate key is a candidate key not selected as the primary key.

Client

Driver_license_number should be an alternate key

It is assumed that each driver license number is unique and each client has exactly one valid driver license number.

D.4 Required Data Constraints

The following attributes should be mandatory and cannot be null.

Outlet

Outlet_number

Address

Phone_number

Vehicle

Registration_number

Make

Model

Engine_size

Capacity

Current_mileage

Daily_hire_rate

Outlet_number

Client

Client_number

First_name

Last_name

Address

Phone_number

DOB

Driver_license_number

Staff

Staff_number

First_name

Last_name

Home_address

Home_phone_number

DOB

Sex

Date_joined

Job_title

Salary

Outlet_number

Hire_Agreement

Hire_number

Client_number

Registration_number

Date_started

Date_wish_end

Mileage_before

Mileage_after may be null at the time a hire agreement is created, because the vehicle may not yet have been returned. Unless they have tracking system can track the current mileage while the client still has it. But it will be costly and unnecessary.

D.5 Attribute Domain Constraints

These constraints define valid values or formats for attributes.

Outlet

Phone_number must contain a valid phone number

Fax_number must contain a valid fax number or may be null if unavailable

Vehicle

Engine_size must be greater than 0

Capacity must be a positive integer

Current_mileage must be greater than or equal to 0

Daily_hire_rate must be greater than 0

Client

DOB must be a valid past date

Driver_license_number must follow a valid license format and be unique

Staff

DOB must be a valid past date

Sex should be limited to an accepted domain such as 'M', 'F', 'Other'

Date_joined must be a valid date

Salary must be greater than 0

Hire_Agreement

Date_started must be a valid date

Date_wish_end must be a valid date

Mileage_before must be greater than or equal to 0

Mileage_after must be greater than or equal to Mileage_before when recorded

D.6 General Constraints

Each vehicle is assigned to exactly one outlet at a given point in time.

Each staff member is assigned to exactly one outlet.

Each hire agreement must involve exactly one client and one vehicle.

A client may have many hire agreements over time.

A vehicle may appear in many hire agreements over time.

The hire end date must not be earlier than the hire start date.

A hire agreement cannot be created unless the referenced client and vehicle already exist in the database.

E.

